

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type	Mammalian, suspension	Molecules Electroporated	DNA: linear, 5 kB
Species Used	Human, Raji, Burkitt lymphoma; Jurkat, acute T cell leukemia; T-cells		

Before the Pulse

Cell Growth Medium	RPMI 1640 + 10% Fetal Calf Serum (GIBCO/BRL, Sigma)	Growth Phase at Harvest	Not given
		Pre-pulse Incubation	None

Wash Solution Hepes Buffered Saline

The Pulse

Instruments Used Gene Pulser® apparatus & Capacitance

Electroporation Temperature	4 °C		
Electroporation Medium*	Phosphate Buffered Saline	Cuvette Gap	0.4 cm
Cell Density	1 x 10 ⁷ (7) cells / pulse	Voltage	0.40 kV
Volume of Cells	0.5 to 0.75 ml	Field Strength	1.0 kV/cm
DNA Concentration	Not given		
DNA Resuspension Buffer	Not given	Capacitor	960 µF
Volume of DNA	11 µl	Resistor	(Pulse Controller) Ω none
After the Pulse		Time Constant	Not given
Outgrowth Medium	Not given		

Outgrowth Temperature	37 °C	Relevant Publications and/or Comments Note: exponential values designated in parentheses. PBS: 1x = 8g NaCl, 0.2g KCl, 0.2g KH ₂ PO ₄ , 1.15g Na ₂ HPO ₄ HBS: 10mM HEPES, pH 7.2, 150 mM NaCl, 5 mM CaCl ₂	
Length of Incubation	2 days		
Selection Method or Assay Used	Preparation of RNA or CAT assay		
Electroporation Efficiency	Not quantified		
Per Cent Survival	50%		

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